

High-Resolution Weather Forecasting Using Dual Input Transformer

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Figure 1: Cloud formation in front of starry sky. via World Meteorological Organization. (<https://wmo.int/topics/weather>).

1 Introduction

Weather forecasts are of central importance for a large number of social and economic sectors, including agriculture, civil protection, energy supply and air traffic. Conventional meteorological forecasts are based on numerical weather prediction models that simulate physical laws. These models are based on scientific findings, have been validated over a long period of time, and offer a high degree of comprehensibility and interpretability. However, they also involve high computing costs, long simulation times, and limited scalability [1, 4].

In contrast, machine-learning-based (ML) approaches offer significant advantages in terms of computational effort, energy efficiency and scalability. Furthermore, current studies show that ML models are comparable to or even surpass physical models in terms of their prediction accuracy, especially for extreme events [1, 4]. Moreover, the open availability of many ML models favours access to research. On the other hand, there are challenges in terms of interpretability and limited integration into operational forecasting systems.

Figure 2 and 3 illustrate the forecast accuracy of different models, which are evaluated using the Root-Mean-Square-Error (RMSE) for the 2-metre temperature and the 10-metre wind speed. The analysis shows that the AIFS model has lower errors than the previous version and even the physical IFS model for both short and long forecast periods.

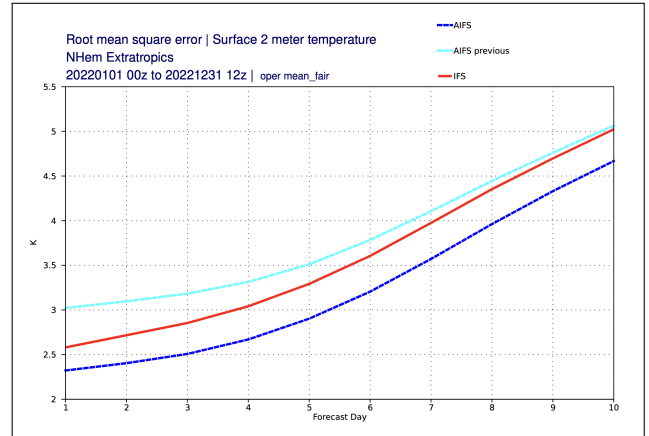


Figure 2: RMSE of 2m temperature for AIFS (blue, dashed), IFS (red, solid) and previous lower resolution AIFS version (cyan, solid) [5].

The aim of this work is to generate high-resolution weather forecasts using a vision transformer (ViT) model with a dual-input structure. In this study, the proposed model is compared with a Convolutional Neural Network (CNN) and a single-input ViT in terms of its prediction performance.

This paper is organised as follows: The second section is devoted to the problem statement, while the third section deals with

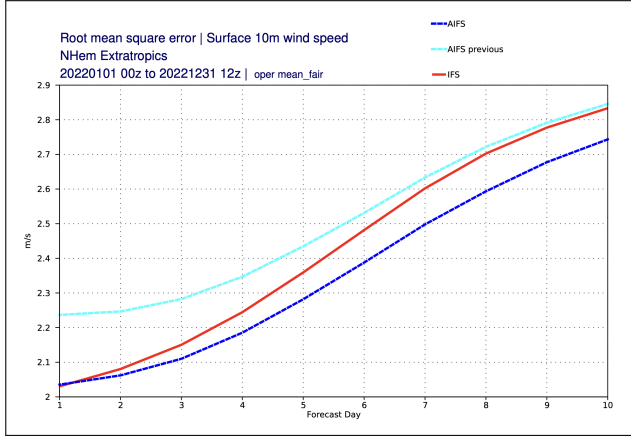


Figure 3: RMSE of 10m wind speed for AIFS (blue, dashed), IFS (red, solid) and previous lower resolution AIFS version (cyan, solid) [5].

the preparatory data processing. Section 4 explains the chosen approach, while section 5 presents the results. Finally, conclusions are drawn in section 6 and potential improvements are discussed in section 7.

2 Problem definition

The precise prediction of meteorological variables in high spatial resolution represents a methodological and modelling challenge. In particular, the short-term forecasting of variables such as the 2m temperature, the horizontal wind components at a height of 10m (U and V) and the relative humidity requires models that can capture both local fine structures and large-scale atmospheric relationships. The focus is on the forecast of a single time step $t+1$ with a horizon of three hours.

A central problem is the integration of a high-resolution target grid with a limited spatial section and contextual information from a larger-scale but lower-resolution environment. The present data structure can be characterised as dual, which places specific demands on the model architecture. On the one hand, the model must capture local details with high accuracy, and on the other hand, it must process sufficient context to correctly categorise mesoscale patterns such as fronts, pressure systems or convection zones.

The challenge is therefore to develop a model that takes into account both cross-scale dependencies and locally accurate predictions. In this context, conventional neural networks, especially convolutional networks, show significant limitations in terms of their spatial generalisation and context integration capacities. Therefore, it is necessary to investigate alternative architectural concepts that can efficiently combine both input types and provide robust predictions on the high-resolution target grid.

3 Theoretical foundations

The application of ML in weather forecasting is based on the data-driven modelling of geophysical processes. In contrast, conventional weather predictions are based on physically founded differential equations, while ML models use historical data to directly

Table 1: Parameter

Parameter	Abbreviation	Unit
Relative humidity	r	%
Temperature	t	$^{\circ}\text{C}$
U-component of wind	u	m s^{-1}
V-component of wind	v	m s^{-1}
Geopotential	z	$\text{m}^2 \text{s}^{-2}$

approximate the relationship between input states and future atmospheric states. In this work, the prediction is formulated as a supervised learning regression problem. Target variables such as temperature, wind or humidity are predicted for a future point in time.

Meteorological data are typically found as regular grids with several physical channels. These can be interpreted as tensors, where each grid point field corresponds to a channel. This interpretation is comparable to multi-channel images in computer vision. One challenge is to adequately capture both small-scale structures (e.g. local temperature gradients) and large-scale meteorological patterns (e.g. frontal systems). In order to do justice to this, it is necessary to use models that are characterised by a high spatial sensitivity and at the same time can access the global context.

CNNs are an established approach for processing such data. They use locally limited filters to extract hierarchical features from image data. However, their inherent limitation to local receptive fields restricts the modelling of broader dependencies. In addition, their ability to integrate context only scales to a limited extent with increasing image size, which is problematic for large-scale meteorological fields.

ViTs represent an alternative architecture that was originally developed for image classification tasks. The underlying architecture is the Transformer architecture, whose core element is the so-called self-attention. This is a mechanism that allows global dependencies between input elements to be modelled. ViTs divide an input image into non-overlapping patches that are processed as a linearised sequence. Each patch is represented by a learnable embedding that is supplemented by position-specific information (positional encoding). The self-attention mechanism allows the model to recognise relationships between all image areas regardless of their spatial distance. This enables simultaneous consideration of both local and global contexts. This results in a significant advantage for meteorological data: large-scale dynamics, such as those occurring in fronts, pressure fields or large-scale convection, can be modelled directly without having to rely on implicit aggregations through convolution layers.

4 Data preprocessing

For the training and evaluation of the model, meteorological re-analysis data from two different sources were used: (1) the CERRA (Copernicus European Regional ReAnalysis) sub-daily regional re-analysis data for Europe on pressure levels [3] and (2) the ERA-5 hourly data on pressure levels [2], both provided by the Copernicus Climate Data Store (CDS). The selected period covers a period of 36 months, starting on 01/2019 and ending on 12/2021.

The selection of meteorological variables comprises five parameters: relative humidity, temperature, U-component of wind, V-component of wind and geopotential (Table 1). These ensure the maximum joint data availability of both data sets for a three-hour time interval in the selected period. The vertical resolution is based on seven pressure levels in hectopascals: 1000, 925, 850, 700, 500, 300 and 200. This selection reflects the vertical data availability. Lower pressure levels (higher altitude) are more sparsely sampled, while there is a higher resolution in layers close to the ground.

For the ERA-5 data, data was obtained monthly via the official CDS API. The extracted data comprises a $32^{\circ}32'$ grid with a horizontal resolution of 31,0 km (total area: $992^{\circ}992'$ km). The original dimensional structure ($32^{\circ}32'5^{\circ}7'$) was merged into a single dataset for the entire 36 months after conversion from GRIB to NetCDF format. A reshaping into the form ($32^{\circ}32'35'$) was then performed, where 35 represents the number of parameter-pressure level combinations. As part of the normalisation process, the mean value and standard deviation are calculated and saved for each parameter-pressure level combination.

The CERRA data was obtained monthly via the CDS API. After completing the initial download process for the entire European region, the grid parameters in the configuration file were manually adjusted to reduce the resolution to $16^{\circ}16'$ cells with a resolution of 5,5 km. This corresponds to a section of $88^{\circ}88'$ km. A reprojective conversion from the Lambert conformal conic projection to a regular latitude/longitude grid was carried out to ensure standardisation of the projection of the CERRA and ERA-5 data. Subsequently, the same process was carried out as for ERA5. A conversion to NetCDF was then carried out, followed by a merge transformation and a reshaping operation to a structure of ($16^{\circ}16'35'$). The data was then normalised.

Finally, a train/validation split was performed over the 36-month period from 01/2019 to 12/2021: five months were randomly selected for validation, while the remaining 31 months were used for training.

5 Approach

The following section presents the approach developed for creating high-resolution weather forecasts based on machine learning methods. In addition to the model architecture, central design decisions, training details and hyperparameter tuning are explained in order to transparently illustrate the methodological principles and the reproducibility of the procedure.

5.1 Objective

As part of this project, the weather forecasting task was formulated as a supervised regression problem. The aim is to predict meteorological fields for the subsequent time $t+1$ based on a given atmospheric state at time t , using a time interval of three hours. This approach is characterised by a continuous prediction of several physical variables. This is in contrast to classical classification approaches, in which a discrete prediction of individual variables is made. Instead, the use of regressive methods is required to enable continuous prediction.

5.2 Input and output representation

The input consists of atmospheric fields of the reanalysis data sets CERRA ($16^{\circ}16'35'$) and ERA-5 ($32^{\circ}32'35'$) for a given time t . Depending on the model variant, either only CERRA data (single input) or a combination of CERRA and ERA-5 data (dual input) is used for modelling.

The model generates a tensor ($16^{\circ}16'35'$), which contains a continuous prediction of all meteorological parameters at all pressure levels for each grid cell in the high-resolution target area. Accordingly, each output component represents a physically interpretable value, such as temperature or wind speed.

A particular challenge is the consideration of spatio-temporal dependencies. The forecast is made exclusively on the basis of individual temporal sections (no sequential input), while the spatial structure of the atmospheric fields plays a decisive role in the quality of the modelling. The chosen modelling architecture must therefore be able to capture both local and large-scale patterns and project them onto the high-resolution target area.

5.3 Model architecture

Three different model architectures were implemented as part of the project. The selection of these models is based on complementary competences in the representation of spatial structures and the ability to integrate large-scale context information. The CNN was chosen as the initial model because it is established in many applications for pattern recognition and enables efficient extraction of local features due to its locally restricted, translationally invariant filter structures. It was found that CNNs are highly suitable for modelling spatially correlated data with a fixed grid structure. A disadvantage, however, is the limited ability to capture long-range dependencies, as these can typically only be represented by deep architectures with many convolutional layers, which increases the model complexity. The architecture used acts as a base model for comparison with transformer-based approaches and processes only the CERRA input tensor. The architecture of the neural network comprises an input layer, five successive convolutional hidden layers and a final output layer for regression prediction. In the hidden layers, ReLU activation functions, batch normalisation and same padding are used to maintain the spatial dimensions. The architecture comprises around 4.2 million trainable parameters and was deliberately dimensioned in terms of its capacity so that it is comparable with transformer-based models. This allows for a comparison that is as fair and efficient as possible in terms of model complexity, enabling a balanced evaluation of performance between the architectures.

The single input ViT transfers the Transformer principle to the processing of two-dimensional grid data. This architecture is characterised by the fact that, in contrast to CNNs, it can model global dependencies directly. This proves to be advantageous for atmospheric processes that are characterised by large-scale dynamics. The single-input ViT architecture is based exclusively on the high-resolution CERRA data as input. Firstly, the input field is segmented into flat patches, which are then processed as a sequence. A separate patch embedding is calculated for each of the 35 variable dimensions. In addition, sine-cosine based position embeddings are used to explicitly encode the spatial position of the patches

in the model. In addition, the variable channels are embedded using so-called channel embedding, which differentiates between various physical parameters. A key feature of this architecture is the integration of temporal information. The lead time interval of the forecast is projected into the same embedding space via the so-called timestep embedder and added to the patch token embeddings, enabling the model to implicitly distinguish between different forecast horizons. The central encoder block is based on the standard nn.TransformerEncoder module and comprises five consecutive encoder layers, each with four self-attention heads (multi-head attention). The feed-forward network within each layer has a hidden dimension four times larger than the embedding dimension. The GELU method, which has proven to be effective in transformers, is used as the activation function. For prediction, the output tokens are passed through an MLP head that maps them back to patch-level values. These reconstructed patches are then converted back to the original input format (a 16×16 grid) using an unpatchify operation.

The dual-input ViT extends the single-input architecture by incorporating a parallel branch for low-resolution ERA5 data, introducing additional large-scale context into the model. This design explicitly addresses the challenge of combining high-resolution local forecasts with synoptic-scale information. While the dual-input setup increases model complexity, it enhances the ability to capture both local and global structures efficiently.

Conceptually, the dual-input ViT builds upon the single-input architecture but adds a separate input pathway for contextual information from ERA5. The two reanalysis sources—CERRA and ERA5—are processed independently, with dedicated patch embedding modules adapted to their respective spatial resolutions by allowing the use of different patch-sizes. This allows for a sensible representation of both data sources. As in the single-input version, sine-cosine positional encodings and channel embeddings are applied to distinguish spatial positions and physical variables. Temporal information is integrated via a shared timestep embedding, which is linearly projected into the embedding space and added to the patch tokens of both sources. Finally, the resulting token sequences from CERRA and ERA5 are concatenated along the sequence dimension to form a unified input for the transformer encoder, enabling joint processing of fine- and coarse-resolution information.

Despite the use of dual inputs, the model is designed to predict only the high-resolution CERRA data, using an output head identical to that of the single-input model.

5.4 Design decisions

When designing the dual input ViT, particular attention was paid to selecting suitable patch sizes for the two input streams. A patch size of 1×1 was selected for the high-resolution CERRA data, while the low-resolution ERA-5 data is processed with a patch size of 4×4 . This decision is based on the consideration of extracting as fine-grained information as possible from the CERRA data, as it represents the actual target area with high resolution. In this case, a larger patch format could potentially smooth out valuable local structures and thus impair the prediction quality.

For the ERA-5 data, on the other hand, the focus is not on the precise reconstruction of local details, but on providing a rough

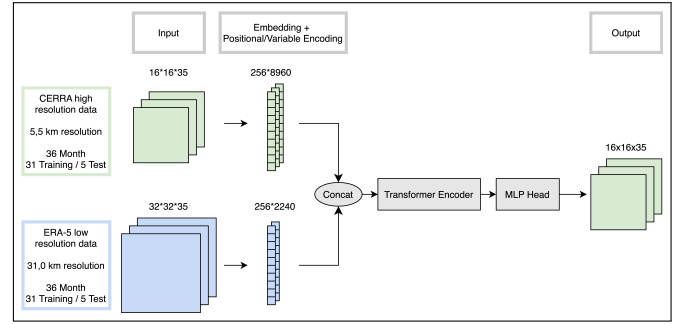


Figure 4: Vision transformer architecture

contextual framework. The use of larger patches in this case is unproblematic in terms of information loss, but significantly reduces memory requirements and computational complexity during training.

In principle, it would also be possible to process the input fields without prior patch formation. In practice, however, this proves to be inefficient due to the quadratically growing memory requirements. The chosen patching strategy therefore represents a compromise between information content and computational effort. A positive side effect of the fine CERRA patch structure is the possibility of returning the model prediction directly to the original grid. This avoids visually disturbing artefacts caused by patch boundaries.

Sine-cosine embedding was used for position coding, which has proven to be robust and efficient in previous work, for example in the Stormer embedding module [6]. This method allows the model to reliably capture relative and absolute position information of the patches in the input grid.

The number of encoder layers was deliberately limited to five. In contrast to long-term time series analyses, for which deep models are necessary, the problem at hand concentrates on a short time horizon and on local spatial relationships. A greater model depth would have led to increased computational and memory requirements, particularly due to the quadratic complexity of the self-attention mechanisms. In addition, an increased risk of overfitting can be observed with increasing depth, especially with a limited training data set.

The dual-input approach represents a key innovation. The attempt to link local fine structures with higher-level dynamic patterns is made by the parallel use of high-resolution CERRA data and coarse-meshed but large-scale ERA-5 data. Although exclusive use of CERRA would be possible for this context, this approach resulted in a significant increase in computational load and storage requirements.

5.5 Training details

The training took place over a period of around 24 hours. In this case, the Adam Optimiser was used as the optimisation method, the performance and stability of which has been proven in particular for deep neural networks with transformer architecture. Adam combines the advantages of adaptive learning rates with

momentum-based updates and thus enables efficient convergence even with complex model structures.

The MSE (F.mse_loss), which is standard in the field of regression problems, was used as the loss function. This choice is particularly suitable as continuous physical variables such as temperature, wind speed or humidity are to be predicted in weather forecasting.

As part of the training process, both the training loss (train_loss) and the validation loss (val_loss) were monitored automatically. The validation loss was calculated after each epoch in order to evaluate the model behaviour on independent data and to identify potential overfitting at an early stage. This monitoring also served as a basis for the subsequent selection of the final model.

6 Results

6.1 Model comparison

As part of the model quality evaluation, a comparison was made between a CNN, a single input ViT, and a dual input ViT in terms of their performance with a comparable number of parameters. A thorough examination of the validation metrics (MSE and RMSE) reveals that the dual-input ViT attains optimal prediction outcomes. It has been demonstrated that this model exhibits the lowest validation loss and produces the most accurate predictions. The CNN is demonstrably capable of generating plausible predictions; however, structural inconsistencies do occur. For instance, localised temperature fluctuations of considerable magnitude occur over brief distances, the geophysical plausibility of which can be categorised as minimal. The presence of artifacts of this nature is indicative of an impaired capacity to accurately capture spatial and contextual patterns.

The transition from CNN to ViT architecture, even when using only single input, results in a significant improvement in model quality. The ViT has been shown to identify spatio-temporal relationships in the input data with greater precision and to project them onto target variables. The resulting forecasts demonstrate a more consistent and meteorologically realistic state for time $t+1$, despite the absence of exact congruence with the ground truth. It is evident that a more substantial enhancement in performance is attained through the implementation of the dual-input ViT. The additional large-scale ERA-5 data appear to provide crucial contextual information that enables the model to better capture the dynamics of weather events. Notably, for weather conditions not encountered during training, forecast errors on the validation set are significantly smaller. This finding indicates that the model has the capacity to extract salient features from coarse-resolution data and effectively transfer them to the finer-resolution forecast.

6.2 Discussion of model differences

Despite the comparable number of parameters exhibited by all models, a substantial disparity in functional complexity is evident. While CNNs process input data in a manner analogous to images and thus tend to learn local patterns, ViTs with their self-attention mechanisms can also capture longer-range and contextual dependencies. This phenomenon elucidates the superior performance exhibited by Transformer-based models. The disparity in performance between the single-input and dual-input ViT can be primarily attributed to the scope and quality of the available information. It is evident

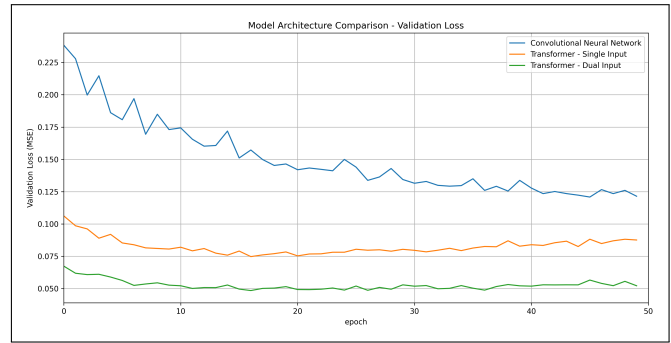


Figure 5: Comparison of validation loss for different model architectures

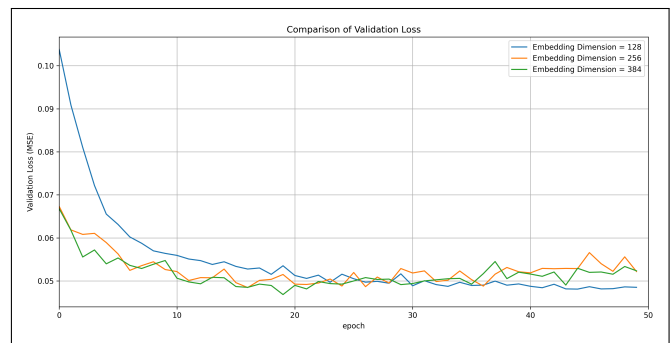


Figure 6: Comparison of validation loss for different embedding dimensions in dual input ViT

that weather patterns are not solely determined by local conditions; rather, they are influenced by large-scale processes. Consequently, forecasts based exclusively on local data are inherently limited in their scope and predictive ability. A comparison of the two models indicates that the incorporation of additional global context information is imperative for ensuring the reliability and precision of predictions.

6.3 Hyperparameter tuning

As part of hyperparameter tuning, the learning rate and embedding dimension were examined as key hyperparameters of the final model. The learning rate controls the step size with which the model weights are adjusted during training, while the embedding dimension determines the size of the vector space in which tokens are represented. Both parameters can significantly influence training dynamics and model quality.

To analyse the influence of the embedding dimension, the model was trained with dimensions of 128, 256 and 384 over 50 epochs. The differences in model performance were smaller than expected and were in a similar range.

In the case of the learning rate, the expected behaviour was observed: higher learning rates resulted in accelerated convergence, but only up to a critical threshold. Once this value was exceeded, the model was no longer able to make meaningful weight updates and

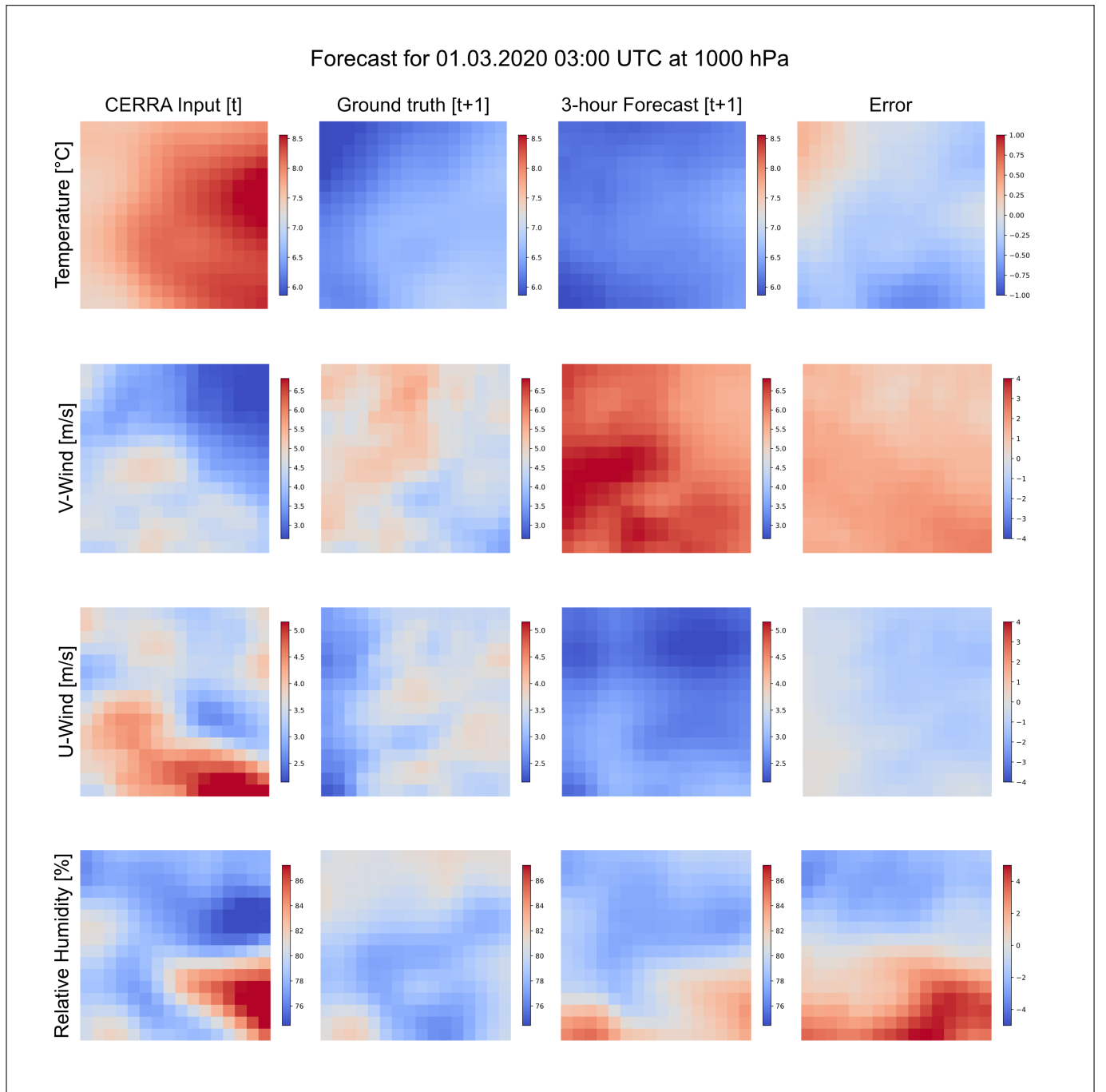


Figure 7: Forecast results of the dual input ViT: Visualisation of input, ground truth, prediction, and error map.

no longer learned effectively. However, no significant differences in MSE were observed within this stable range.

The influence of different patch sizes on model behaviour and output quality was also investigated during the project. Patch size is a key hyperparameter in transformer-based weather forecasting models, as it determines the tokenisation of the input data. During

segmentation, the input field is divided into patches of equal size that do not overlap, with each patch entering the model as a single token. The experiments showed that reducing the patch size has a significant impact on training time. A reduction in patch size results in a fourfold increase in the number of tokens (twofold compression in two spatial dimensions) and thus a corresponding increase in

computational load. The initial patch size was set to 4 and then gradually reduced to 2 and finally to 1 after extensive optimisation of the remaining hyperparameters. This resulted in a noticeable increase in training time under otherwise unchanged conditions. With regard to the validation metric, only marginal improvements were observed with smaller patch sizes, which were not statistically significant. The primary motivation for reducing the patch size was therefore less about numerical optimisation and more about visually improving the quality of the output. In particular, the aim was to reduce the occurrence of patch artefacts i.e. artificially generated, grid-like discontinuities at the patch boundaries. A visual examination of the model predictions confirmed that smaller patches tend to result in a smoother and more consistent spatial structure.

7 Conclusion

This paper demonstrates the effectiveness of a dual input ViT for high-resolution, short-term weather forecasts using limited high-resolution reanalysis data. Compared to a conventional CNN and a single input ViT, the dual input model achieves significantly better prediction results, although all models have a comparable number of parameters. This result emphasises the relevance of additional contextual information from large-scale, low-resolution data sources such as ERA-5, which, in combination with high-resolution CERRA data, lead to a significant increase in prediction accuracy. The integration of the two data sources allows the model to capture relevant structures and generate more precise forecasts at the local level.

Another finding concerns the design of the patch structure in the Vision Transformer. The use of asymmetric patch sizes for CERRA and ERA-5 data enabled a successful reduction of artefacts in the prediction maps, in particular rectangular and erratic patterns, without any significant loss of performance being observed. This suggests that an adaptive representation of different resolutions is not only a pragmatic solution, but also has a positive impact on the quality of the model.

The central objective of the project, the creation of high-resolution weather forecasts for small regions using limited amounts of data, was successfully achieved overall. The results of the study suggest that by using suitable modelling architectures and intelligent data integration, powerful forecasting systems can be implemented even with limited resources.

8 Further improvements

Despite the results achieved, there are several starting points for further development of the model, particularly with regard to data scope, model architecture and application perspectives. A major limiting factor in the course of the project to date has been the limited availability of reanalysis data. Due to the significant duration of the download times and the limited capacity of the storage medium, the scope of the training data set was limited to a period of just three years. However, it is to be expected that the inclusion of larger amounts of data, for example by extending to earlier decades or additional meteorological variables, could significantly improve the generalisation capability of the model.

There is also potential for optimisation in the area of hyperparameter tuning. The parameter studies carried out to date have been limited for resource reasons. A more systematic exploration,

for example using Bayesian optimisation, could lead to an improved configuration of the model. The choice of patch size and its combinations in the ViT represent a significant factor for model performance, especially with regard to the integration of multiscale information.

A promising development path is the extension of the model to multi-step forecasting. At present, the model is limited to the prediction of a single time step ($t+1$). However, the use of so-called rollout methods could generate successive predictions for longer periods of time, which would significantly increase the practical benefits.

The long-term goal is to implement the developed model in an operational forecasting system. The realisation of this measure requires access to current, real-time observation data as well as an efficient architecture for real-time processing. However, the latter is not yet fully realisable.

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